Metadata Assessment

Overview

This document evaluates the metadata quality and documentation completeness of the Dryad dataset associated with the study *"A Study of Impulsivity and Adverse Childhood Experiences in a Population Health Setting."* The assessment was conducted as part of a CURATE(D)-based data curation workflow to determine whether the dataset supports independent understanding, reuse, transparency, and long-term preservation.

The dataset contains participant-level tabular data related to adverse childhood experiences (ACEs), impulsivity indicators, demographic attributes, and associated behavioral measures. Metadata assessment focused on the clarity, completeness, consistency, and interpretability of the deposited files and accompanying documentation.

Metadata Components Reviewed

The following metadata elements were reviewed during the assessment process:

- Dataset title and repository description
- Author and publication information
- DOI and persistent identifier support
- File naming conventions
- Variable naming consistency
- README documentation
- Data structure descriptions
- Column-level interpretability
- Dataset-to-article linkage
- Repository-level descriptive metadata

Observations

1. Repository-Level Metadata

The Dryad repository provides strong basic descriptive metadata including:

- dataset title,
- author information,
- publication linkage,
- DOI assignment,
- repository citation,
- and publication year.

This supports discoverability and long-term identification of the dataset.

Strengths

- Persistent DOI provided
- Public accessibility
- Clear linkage to associated journal article
- Standard repository structure

Limitations

- Limited contextual explanation for dataset reuse
- Minimal methodological guidance for independent users

2. File-Level Metadata

The dataset files are organized in CSV format and are compatible with long-term preservation practices due to their non-proprietary structure.

Strengths

- Machine-readable format
- Consistent tabular organization
- Structured rows and columns

Limitations

- Limited explanation of relationships between files
- Insufficient descriptions of derived variables
- Missing clarification for some abbreviated column names

3. Variable-Level Documentation

Several variables contain abbreviated or domain-specific naming conventions that may be difficult for independent researchers to interpret without consulting the associated article.

Observed Issues

- Some variables lack direct definitions
- Scoring logic for calculated fields is not fully explained
- Missing descriptions for certain coded values
- Limited contextual information for behavioral indicators

These gaps reduce independent interpretability and increase dependency on the associated publication.

4. README Documentation Assessment

The repository includes basic documentation; however, the README information is limited for long-term reuse purposes.

Missing Elements

- Detailed variable descriptions
- Data preprocessing explanations
- Workflow summaries
- File relationship explanations
- Reuse guidance for external researchers

As a result, users may face difficulty understanding the dataset without prior domain knowledge.

Recommended Improvements

The following improvements would significantly strengthen metadata quality and dataset reuse:

1. Add a detailed data dictionary describing all variables and coded values.
2. Expand README documentation with workflow and preprocessing details.
3. Include explanations for derived variables and scoring methods.
4. Provide clearer descriptions of relationships between files.
5. Add reuse guidance for researchers unfamiliar with the study domain.
6. Include versioning or update history for future repository maintenance.

Conclusion

Overall, the dataset provides strong repository-level metadata and preservation support through DOI assignment and open repository access. However, variable-level documentation and contextual explanations remain limited. While the dataset is accessible and structured appropriately for preservation, additional metadata improvements would significantly improve independent understanding, transparency, and long-term reuse.